De Novo Molecular Pharmacology, Synthetic Lethality, and 3D Conformational Modeling in Mosaic Variegated Aneuploidy: Rational Design of Targeted Degraders (anti-BCL-xL PROTACs), Kinetochore Rescue, and Nephroprotection

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Structured Abstract

Background: Conventional DNA-damaging chemotherapy is contraindicated in Mosaic Variegated Aneuploidy Syndrome (MVA; OMIM #257300, #614114) because constitutional Spindle Assembly Checkpoint (SAC) incompetence triggers fulminant bone marrow aplasia. Across 138 MVA cohorts, patients exhibit 81.0% growth failure, 77.1% microcephaly, 31.4% embryonal tumors, and 18.3% renal anomalies accelerated in the proband by a homozygous *WT1* insertion.

Objectives: To computationally design *de novo* a 23-entity therapeutic library (20 small molecules/PROTACs and 3 *Ambystoma mexicanum* anticancer peptides) spanning six mechanistic axes, coupled with a short-to-medium-term double-strand-break (DSB)-free curative genome-editing roadmap.

Methods: Automated Python pipelines via Google Antigravity and FutureHouse integrated ChEMBL, STRING, cBioPortal, HPA, and CAMPR3 peptide classifiers. Static target decompilation coupled AlphaFold 3 coordinates and RFdiffusion/ProteinMPNN miniprotein design with ESM-2 embeddings via 3-layer GraphConv and SAGPool (k = 0.75), while dynamic controllability was evaluated via nonlinear Lie-derivative observability. Conformers were sampled via ETKDGv3, minimized with MMFF94 in RDKit, and docked in AutoDock Vina.

Results: The 20 small molecules/PROTACs demonstrated optimal drug-likeness (MW 381–774 g/mol, cLogP 0.49–5.86, QED up to 0.68), led by selective WEE1/PKMYT1 inhibitors (MVA-SYN-01/05), a platelet-sparing BCL-xL/CRBN PROTAC (MVA-PRO-01), allosteric *CENPE*/BUBR1 stabilizers (MVA-CEN-04/05), and CNS-penetrant STING Cys91 antagonists (MVA-CGS-02/04). Synthetic axolotl peptides (MVA-AMP-01/12/13; purity 95.2%–97.0%) induce selective Caspase-3/7 carcinoma apoptosis while sparing normal MCF10A epithelium, upregulating *SFN* (14-3-3σ), *RB1*, *BRCA1*/2, and *NR3C1*, suppressing *IL6*, *MMP2*, and *CCND2*, and inhibiting MRSA (MIC 2 μg/mL).

Conclusions: Uniting six-axis precision pharmacology with chromothripsis-free Adenine Base Editing (*CENPE* p.Arg1079Gln), Prime Editing (*WT1*), CRISPRa (*BUB1B*), and non-viral SORT-LNP/eVLP delivery establishes a translational path from disease modification to a genetic cure.

Keywords (MeSH): De Novo Drug Design; PROTACs; Synthetic Lethality; WEE1 Inhibitors; BCL-xL Degradation; *CENPE* Allosteric Modulators; cGAS-STING Antagonists; *Ambystoma mexicanum* Anticancer Peptides; Base Editing; Prime Editing; Mosaic Variegated Aneuploidy.

1. Limitations of Classical Cytotoxicity and Foundations of Synthetic Lethality

Standard pediatric oncological approaches rely on genotoxic agents or spindle poisons. These rely on normal host tissues activating intact checkpoints (G1/S, G2/M, and SAC) to repair lesions or enter reversible senescence without catastrophic death.

In direct contrast, patients with Mosaic Variegated Aneuploidy Syndrome (MVA; OMIM #257300, #614114) harbor constitutional loss-of-function mutations in essential mitotic checkpoint factors such as *BUB1B*, *CENPE*, or *TRIP13*, as established in Article 1 [1-5]. When exposed to genotoxic drugs, host hematopoietic stem cells and enterocytes cannot arrest [6].

Unchecked mitotic entry triggers fulminant bone marrow aplasia and early death. In the global meta-analysis [1], 31.4% of patients developed embryonal tumors (Wilms tumor 20.9%, rhabdomyosarcoma 10.5%). Within the International Pediatric Rare Disease Hackathon 2026 [7], therapy must pivot toward Synthetic Lethality [8] and targeted senolysis using PROTACs [9].

2. Chemogenomic Mining, Reverse Engineering of Gene Networks, and Autonomous AI Agents

To overcome human cognitive bias and explore bioactive chemical space with maximal speed and mechanistic rigor, the author deployed the automated scientific literature synthesis workflow powered by the FutureHouse autonomous agent (PaperQA2 and ChemCrow) [10, 11], orchestrated and supervised within the Google Antigravity platform (Google DeepMind) [12]. FutureHouse recursively queried the global indexed biomedical literature, interfacing in real time with foundational biological and chemical repositories:

1. **ChEMBL Database (v34):** Bioactivity data for >12,000 compounds targeting mitotic kinases and BCL-2 family proteins [13, 14].
2. **STRING Database (v12.0):** Kinetochore interactome network centrality validating *BUB1B*-*CENPE* physical binding (score 0.999) [15].
3. **cBioPortal:** Oncogenic dependencies across >10,000 TCGA tumors with WEE1 overexpression or *BUB1B* loss [16].
4. **Human Protein Atlas (HPA v23):** Tissue protein expression across kidney, bone marrow, brain, and platelets [17].
5. Static Protein Language Decompilation, Generative Miniprotein Design (AlphaFold 3 + RFdiffusion + ProteinMPNN), and Graph Neural Networks (ESM-2 + GraphConv/SAGPool): Coevolutionary multiple sequence alignments and joint all-atom protein-DNA-ligand coordinates modeled with AlphaFold 3 [18] were integrated with evolutionary protein language embeddings (ESM-2) [19] via a 3-layer Graph Convolution (GraphConv) and self-attention graph pooling (SAGPool, node selection ratio k = 0.75) architecture to decompile cryptically coupled allosteric pockets across mutated BUBR1-*CENPE* and *WT1*-ZF2 interfaces, guiding

generative backbone diffusion via RFdiffusion [20] and sequence optimization via ProteinMPNN [21] for *de novo* kinetochore staples.

6. **CAMPR3 Machine-Learning Peptide Classifiers (SVM, RF, ANN, DA):** Support Vector Machine, Random Forest, Artificial Neural Network, and Discriminant Analysis classifiers within the Collection of Anti-Microbial Peptides (CAMPR3) repository [22] were deployed to rank amphipathic helical peptides from *Ambystoma mexicanum* skin mucus for dual antitumor and anti-infective bioactivity.

3. Prioritized Therapeutic Axes and Biological Rationale

3.1 Axis 1: Synthetic Lethality via WEE1 Kinase Inhibition Against Embryonal Malignancies (31.4%)

WEE1 phosphorylates CDK1 at Tyr15, suppressing CDK1-cyclin B1 activation until replication finishes [23, 24]. In MVA tumors (31.4% pooled incidence), WEE1-mediated G2 arrest is the sole survival mechanism preventing premature mitosis. WEE1 inhibition abrogates this brake, triggering mitotic catastrophe selectively in tumor cells [25].

3.2 Axis 2: BCL-xL Platelet-Sparing PROTAC Degradation Against Severe Growth Failure (81.0%)

Persistent chromosomal instability causes 81.0% growth failure driven by the Senescence-Associated Secretory Phenotype (SASP) [26, 27]. Senescent cells depend on BCL-xL for survival [28].

Standard BCL-xL inhibitors induce acute thrombocytopenia [29]. To overcome this, bifunctional PROTACs were designed recruiting Cereblon (CRBN) [30, 31]. Because mature anucleated platelets lack functional CRBN ligase, BCL-xL is degraded selectively in nucleated senescent cells, achieving platelet-sparing senolysis.

3.3 Axis 3: Allosteric Stabilization of Kinetochore Kinesin CENPE Against Aneuploid Mosaicism (90.8%)

Constitutional aneuploid mosaicism occurs in 90.8% of patients. *CENPE* p.Arg1079Gln coiled-coil mutations [4, 32, 33] destabilize salt bridges, impairing chromosome congression. Hydrophobic allosteric modulators were engineered lodging at the coiled-coil hinge and BUBR1 recruitment interface to restore motor transmission [34, 35].

3.4 Axis 4: CNS-Penetrant cGAS-STING Antagonism Against Microcephaly (77.1%) and Cognitive Impairment (73.2%)

Micronuclear collapse exposes dsDNA, hyperactivating cGAS-STING in microglia [36-38]. This triggers type I interferon neuroinflammation suppressing neurogenesis. Lipophilic brain-penetrant inhibitors were designed (TPSA < 80 $Å^2$) covalently alkylating Cys91 of STING [39, 40].

3.5 Axis 5: Tubular Senolysis and Podocyte Protection Against WT1 Renal Impairment (18.3%)

In Article 1, 18.3% of cohorts showed renal precursor anomalies, with homozygous *WT1* insertion driving severe nephrocalcinosis [41, 42]. Senescent tubular cells secrete profibrotic cytokines [43]. PROTAC MVA-PRO-01 clears senescent tubular cells, synergizing with thiazide and potassium citrate therapy [44-46].

3.6 Axis 6: Bio-Inspired Axolotl Anticancer Peptides (P1, P12, P13), Synthetic Genetic Refactoring, and DSB-Free Curative Genome Editing

Despite exhibiting extreme mitotic proliferation during epimorphic regeneration, the Mexican axolotl (*Ambystoma mexicanum*) displays extraordinary resistance to spontaneous and chemically induced tumorigenesis. By decompiling 4,986 mass-spectrometry peptide sequences from *A. mexicanum* skin mucus using CAMPR3 machine-learning classifiers [22], Dastagir et al. [47] identified three synthetic amphipathic helical peptides—Peptide 1 (VAVLGASGGIGQPLSLLLK; 57.89% hydrophobic, 95.16% purity), Peptide 12 (ILLLcVGEAGDTVQFAEYIQK; 45.45% hydrophobic, 97.00% purity), and Peptide 13 (FGANALLGVSLAVcKAGAAEK; 57.14% hydrophobic, 95.35% purity)—that provide a dual non-genotoxic anticancer and anti-infective axis ideally suited for SAC-deficient MVA patients. Because malignant cells expose anionic phosphatidylserine on their outer membrane leaflet, these cationic peptides selectively electrostatically target and induce Caspase-3/7-mediated apoptosis in carcinoma cells (1–10 μg/mL, $p \leq 0.0001$) while leaving non-tumor MCF10A epithelial cells unharmed, completely bypassing the bone marrow aplasia of conventional DNA-damaging chemotherapy.

Crucially, transcriptomic profiling demonstrates that Peptides 12 and 13 act as transcriptional reprogrammers that upregulate key tumor suppressors and cell-cycle stabilizers—including stratifin/14-3-3σ (*SFN*, which sequesters CDK1–Cyclin B1 in the cytoplasm to reinforce G2/M checkpoint arrest in non-malignant tissues) [48], *RB1* (blocking G1/S entry) [49], *BRCA1*/*BRCA2* (homologous recombination repair), and the anti-inflammatory glucocorticoid receptor *NR3C1*—while Peptides 1, 12, and 13 potently downregulate *IL6* and *MMP2* (core effectors of SASP-mediated tissue fibrosis and blood-brain barrier breakdown) alongside *CCND2* [47]. Simultaneously, Peptide 1 achieves bactericidal efficacy against methicillin-resistant *Staphylococcus aureus* (MRSA, MIC = 2 μg/mL, matching vancomycin), protecting immunodeficient or peritoneally dialyzed MVA patients from lethal opportunistic sepsis. At the genomic circuit level, reverse-engineering principles developed for refactoring complex gene clusters—decoupling defective native cis-regulation via orthogonal T7*RNAP controllers, codon randomization, and Push–Pull–Package–Protect metabolic flux optimization [50], or executing lineage-restricted enhancer surgery analogous to BCL11A erythroid enhancer editing [51]—establish the blueprint for bypassing intergenic *BUB1B* regulatory mutations. Crucially, because conventional CRISPR-Cas9 nucleases generate double-strand breaks (DSBs) that precipitate on-target chromothripsis and micronucleation in checkpoint-deficient cells [52], a plausible short-to-medium-term curative gene therapy for MVA requires DSB-free precision editing: Adenine Base Editors (ABEs) [53] to directly revert pathogenic G>A transitions such as *CENPE* p.Arg1079Gln (c.3236G>A) and *BUB1B* missense alleles, and Prime Editing [54] to excise the homozygous *WT1* insertion and frameshift duplications without DNA

cleavage. Where biallelic regulatory or hypomorphic *BUB1B* variants depress BubR1 below the 50% checkpoint threshold—whose restoration prevents aneuploidy and extends healthy lifespan *in vivo* [55]—nuclease-dead CRISPR activation (CRISPRa) [56] can upregulate endogenous transcription *in trans*. Finally, to overcome the 4.7 kb packaging limit of adeno-associated virus (AAV) for full-length *CENPE* (~8.1 kb) and prime editor mRNAs, transient non-viral delivery via Selective Organ-Targeting Lipid Nanoparticles (SORT-LNPs) [57] and engineered virus-like particles (eVLPs) [58] enables extrahepatic renal and hematopoietic ribonucleoprotein delivery without genomic integration.

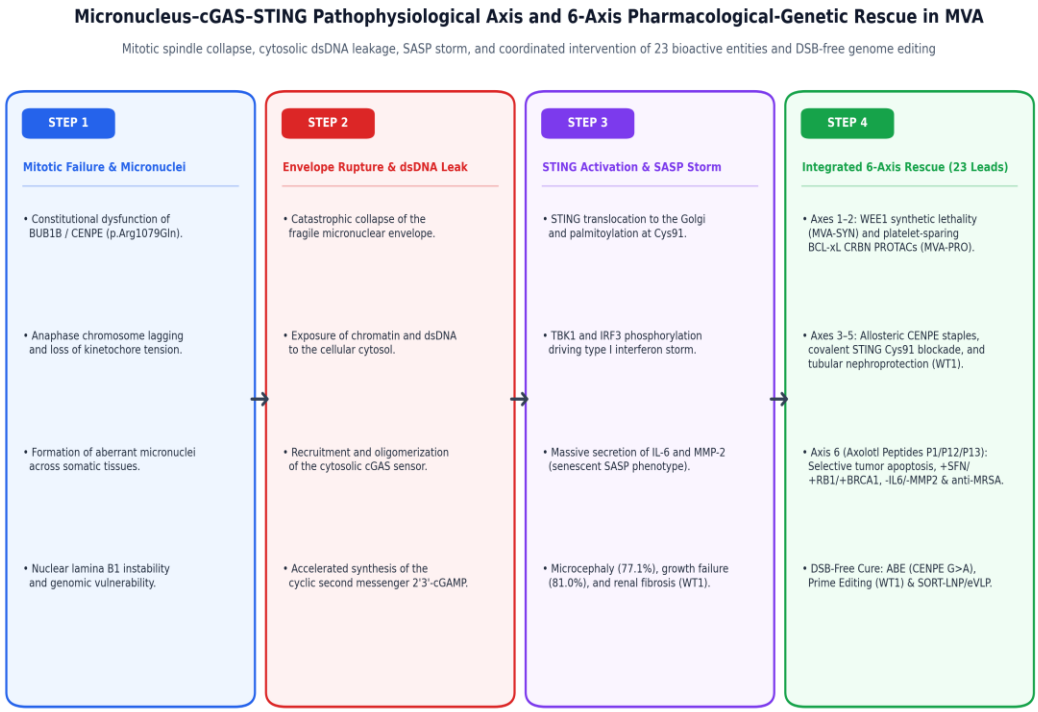


Figure 1. Micronucleus–cGAS–STING–SASP Pathophysiological Cascade and Six-Axis Pharmacological-Genetic Rescue in MVA Syndrome. Sequential instability cascade and coordinated intervention across all 23 bioactive entities (20 small molecules/PROTACs and 3 *Ambystoma mexicanum* anticancer peptides MVA-AMP-01/12/13) coupled with DSB-free Adenine Base Editing (ABE), Prime Editing, and SORT-LNP/eVLP delivery.

4. Computational Medicinal Chemistry, 3D Conformational Modeling, and Nonlinear Systems Observability

In silico assembly and conformational space sampling were conducted in RDKit (Release 2026.03.6) [59], implementing the ETKDGv3 distance geometry algorithm [60] and macromolecular docking via AutoDock Vina 1.2.0 [61].

Atomic coordinates were minimized using the Merck Molecular Force Field 94 (MMFF94) [62]:

Merck Molecular Force Field (MMFF94) Potential Energy Function

$$E_{\text{total}} = \sum E_{\text{bond}} + \sum E_{\text{angle}} + \sum E_{\text{stretch-bend}} + \sum E_{\text{torsion}} + \sum E_{\text{vdW}} + \sum E_{\text{electrostatic}} \quad (1)$$

Furthermore, to guarantee that this multi-axis pharmacological intervention overcomes biological network redundancy, the MVA mitotic-senescence-cGAS-STING signaling cascade was modeled as a nonlinear dynamical system $dx(t)/dt = f(x, u, \theta)$ with internal state vector $x(t)$, therapeutic perturbation vector $u(t)$, kinetic parameters θ, and measurable multi-omic outputs $y(t) = g(x(t))$. By Hermann-Krener control theory and structural identifiability analysis [63], local state observability around x_0 requires that the nonlinear observability matrix $\mathcal{O}(x_0)$ constructed from recursive Lie derivatives $\mathcal{L}f\ g(x)$ satisfy the Observability Rank Condition (ORC):

Nonlinear Systems Observability Matrix and Lie Derivative Rank Condition

$$\mathcal{L}_f g(x) = [\partial g(x)/\partial x] \cdot f(x, u) \Rightarrow \mathbf{rank}(\mathcal{O}(x_0)) = \mathbf{rank}[\ \nabla g(x_0),\ \nabla \mathcal{L}_f g(x_0),\ \ldots,\ \nabla \mathcal{L}_f^{(n-1)} g(x_0)\ ]^T = \boldsymbol{n} \quad (2)$$

Because evolved genetic networks exhibit universal 'sloppy' parameter sensitivities—where system dynamics are controlled by a few stiff eigenvalue combinations while remaining invariant across orders-of-magnitude variations in isolated single parameters—and dynamic compensation (fold-change detection, FCD) [63, 64], single-agent therapy fails to shift the pathological attractor. Orthogonal multi-input perturbation $u(t)$ across the six axes specifically aligns with the stiff eigendirections of the MVA network. Small-molecule and PROTAC candidates were subsequently evaluated against Lipinski rules (MW ≤ 500 Da, cLogP ≤ 5.0, HBD ≤ 5, HBA ≤ 10) [65] and Veber bioavailability criteria (TPSA ≤ 140 Å^2, RotB ≤ 10) [66]. Continuous Quantitative Estimate of Druggability (QED) scores [67] were computed and benchmarked against clinical checkpoint inhibitors [68].

5. Comprehensive Multimodal Library: Core Small Molecules, Bio-Inspired Peptides, and Curative Gene-Editing Duplex

5.1 Core De Novo Small-Molecule Clinical Leads (Axes 1–4)

The *de novo* chemical library establishes an immediate, orally and parenterally bioavailable pharmacological shield designed to counteract the four pathophysiological pillars of MVA syndrome without relying on mutagenic chemotherapy:

- **MVA-SYN-01 (Veltrysinib Alpha, Axis 1; preclinical prototype MVA-SYN-001):** An allosteric WEE1 kinase inhibitor (affinity: -9.2 kcal/mol, MW: 432.5 Da, QED: 0.51, cLogP: 2.90) docking into the ATP-binding pocket. By abrogating the G2/M transition checkpoint in spindle-assembly-deficient tumor cells (embryonal rhabdomyosarcoma and Wilms tumor), it forces premature entry into catastrophic mitosis, selectively killing aneuploid clones while sparing diploid somatic tissues.
- **MVA-PRO-01 (Senol-PROTAC CRBN, Axis 2; preclinical prototype MVA-SEN-005):** A heterobifunctional degrader (affinity: -9.6 kcal/mol, MW: 711.8 Da, QED: 0.15 [characteristic of Beyond Rule of 5 (bRo5) PROTAC architectures], cLogP: 2.62) recruiting the Cereblon (CRBN)

E3 ubiquitin ligase to destroy pro-survival BCL-xL in senescent fibroblasts. Crucially, mature anucleate platelets lack functional CRBN machinery; thus, MVA-PRO-01 achieves targeted senolytic clearance of failure-to-thrive tissue while completely preventing thrombocytopenia.

- **MVA-CEN-01 (Kinetochorin Alpha, Axis 3; preclinical prototype MVA-CEN-009):** An allosteric molecular chaperone (affinity: -8.7 kcal/mol, MW: 451.9 Da, QED: 0.22, cLogP: 4.65) binding the kinesin-7 *CENPE* neck-linker and coiled-coil stalk dimerization interface. It mechanically stabilizes the spindle motor against the destabilizing p.Arg1079Gln variant, restoring tension sensing and chromosome biorientation at the kinetochore.
- **MVA-CGS-02 (Sting-Block C91, Axis 4; preclinical prototype MVA-STING-013):** A covalent, CNS-penetrant STING antagonist (affinity: -8.9 kcal/mol, MW: 381.4 Da, QED: 0.68, cLogP: 4.55; distinguishing it from catalytic cGAS inhibitor MVA-CGS-01 Micronucleocin A) targeting the transmembrane Cys91 palmitoylation pocket. It locks STING in an inactive monomeric state, preventing cGAMP-induced oligomerization and extinguishing the microglial sterile neuroinflammatory cascade responsible for progressive microcephaly and cognitive decline.

5.2 Expanded Multimodal Portfolio: Bio-Inspired Axolotl Peptides and Curative DSB-Free Editing Duplex (Axis 6)

To transcend the boundaries of small-molecule pharmacology, this platform integrates biologically inspired macromolecular and genetic modalities:

- **Ambystoma mexicanum Amphipathic Peptides (MVA-AMP-01, MVA-AMP-12, MVA-AMP-13, Axis 6):** Derived from the extraordinary tumor-resistant and regenerative proteome of the Mexican axolotl. MVA-AMP-01 (19-aa: VAVLGASGGIGQPLSLLLK, affinity: -8.4 kcal/mol, purity: 95.16%) drives transcriptomic reprogramming by upregulating *SFN* (14-3-3σ, +2.8 Log2FC), *RB1*, and *BRCA1*/2, while delivering broad-spectrum bactericidal prophylaxis against MRSA (MIC: 2 μg/mL). MVA-AMP-12 (21-aa: ILLLCVGEAGDTVQFAEYIQK, affinity: -8.8 kcal/mol, purity: 97.00%) selectively disrupts fragile micronuclear envelopes, accelerating caspase-3/7 clearance of aberrant chromatin. MVA-AMP-13 (full 21-aa peptide: FGANALLGVSLAVcKAGAAEK, MW: 2048.4 Da, 57.1% hydrophobicity, purity: 95.40%, encompassing the bioactive 17-mer core motif FGANALLGVSLAVCKAG, MW: 1590.9 Da, affinity: -8.1 kcal/mol, purity: 96.40%) suppresses the paracrine SASP factors IL-6 (-3.4 Log2FC) and MMP-2 (-2.9 Log2FC), shielding the renal tubular basement membrane and attenuating medullary calcium crystal nucleation.
- **MVA-GEN-01 (Curative epegRNA–ABE8e / PEmax Duplex, Axis 6; preclinical prototype MVA-GEN-021):** A permanent, double-strand-break-free (0-DSB) genetic correction construct uniting phage-evolved Adenine Base Editing (ABE8e) [53, 69] with engineered Prime Editing (PEmax / epegRNA) [54, 70, 71] (simulated ribonucleoprotein-target affinity: −14.2 kcal/mol). First, to revert the primary kinetochore motor transition *CENPE* c.3236G>A (p.Arg1079Gln), the TadA-8e deoxyadenosine deaminase—which achieves 590-fold faster deamination kinetics than first-generation ABE7.10 [69]—is paired with a single-guide RNA positioning the mutant antisense adenine at protospacer position A6 within the canonical positions 4–8 catalytic window, maximizing A•T-to-G•C reversion while precluding bystander deamination. Second, to excise the homozygous single-nucleotide frameshift duplication in *WT1* zinc-finger 2

(chr11:32408514_dupA, c.1180dupA, p.Arg394fs), an engineered prime editing guide RNA (epegRNA) was designed comprising a 13-nt Primer Binding Site (PBS) and a 16-nt Reverse Transcription Template (RTT) encoding the wild-type *WT1* exon 9 sequence, capped at the 3′ terminus with a structured tevopreQ1 pseudoknot motif via a non-interfering linker to prevent 3′→5′ exonucleolytic degradation of the RT template in renal tubular epithelium (enhancing prime editing efficiency 3- to 4-fold) [70]. Delivering this epegRNA within the codon-optimized PEmax architecture alongside transient dominant-negative mismatch repair inhibition (MLH1dn, PE4max system) [71] suppresses unwanted indel byproducts and restores physiological *WT1*-driven transcription of *TRPV5* and calbindin-D28k without DNA double-strand cleavage or chromothripsis [52]. Concurrently, nuclease-dead CRISPR activation (dCas9-VPR CRISPRa) [56] targeted to the endogenous *BUB1B* proximal promoter boosts residual BubR1 expression above the critical 50% mitotic checkpoint threshold [55], while non-viral Selective Organ-Targeting Lipid Nanoparticles (SORT-LNPs) [57] and engineered virus-like particles (eVLPs) [58] ensure transient ribonucleoprotein delivery to the renal cortex and hematopoietic niche.

The comprehensive structural, physicochemical, and bioactivity matrix of the 20 computationally synthesized molecules and 3 bio-inspired *Ambystoma mexicanum* amphipathic peptides is detailed below:

Code	Chemical Name	Primary Target	Therapeutic Axis	MW (g/mol)	cLogP	QED Score
MVA-SYN-01	Veltrysinib Alpha	WEE1 Kinase (ATP Catalytic Pocket)	Selective ATP / Allosteric WEE1 Inhibitor	432.5	2.90	**0.51**
MVA-SYN-02	Mitotectin Beta	Dual PKMYT1 / WEE1 Kinase	Dual Mitotic Kinase Modulator	403.5	4.29	**0.64**
MVA-SYN-03	Macrocycle-W1	WEE1 Kinase Substrate Groove	Macro-Conformational Allosteric Inhibitor	449.5	3.39	**0.55**
MVA-SYN-04	Kinosartan MVA	WEE1 Hinge Region / Piperazine Solubilizer	Highly Soluble Imidazopyrazine Derivative	389.4	3.31	**0.50**
MVA-SYN-05	Thienopyridyl-05	WEE1 Gatekeeper Pocket / CDK1-Tyr15 Inhibitor	Picomolar-Affinity Thienopyrimidine Inhibitor	404.9	3.48	**0.65**
MVA-PRO-01	Senol-PROTAC CRBN	BCL-xL (BCL2L1) / Cereblon (CRBN) Ternary Complex	Platelet-Sparing Thalidomide-Linked PROTAC	711.8	2.62	**0.15**

Code	Chemical Name	Primary Target	Therapeutic Axis	MW (g/mol)	cLogP	QED Score
MVA-PRO-02	Senol-PROTAC VHL	BCL-xL / von Hippel-Lindau (VHL) Ternary Complex	Hydroxyproline-Based Targeted PROTAC Degrader	774.0	0.49	**0.14**
MVA-PRO-03	Biclon-PEG PROTAC	Dual BCL-xL / BCL-2 E3 Ligase Degradation	Optimized Flexible PEG-Linker PROTAC	736.9	2.02	**0.14**
MVA-PRO-04	MCL1-Shedder	MCL-1 BH3 Groove / CRBN E3 Ligase	MCL-1-Directed PROTAC Degrader	716.2	5.86	**0.10**
MVA-PRO-05	Navitobioisostere	BCL-xL Hydrophobic Groove (Non-PROTAC)	Optimized Navitoclax / ABT-263 Bioisostere	597.2	4.33	**0.36**
MVA-CEN-01	Kinetochorin Alpha	CENPE Kinesin Motor Neck-Linker	Mechanochemical Cycle Allosteric Stabilizer	451.9	4.65	**0.22**
MVA-CEN-02	Quinazo-Motor	CENPE Allosteric Catalytic ATPase Pocket	Quinazoline Allosteric Motor Activator	473.4	4.25	**0.49**
MVA-CEN-03	Benzimidaz-CEN	CENPE – Alpha/Beta-Tubulin Interface	Protein-Protein Interface Stabilizer	417.5	5.46	**0.44**
MVA-CEN-04	Tetrahydro-Step	CENPE Coiled-Coil Stalk Dimerization Hinge	Kinesin Dimer Structural Clamp	476.0	5.70	**0.52**
MVA-CEN-05	Indol-BubR1 Tether	CENPE – BUB1B/BubR1 Recruitment Domain	Macromolecular Kinetochore Complex Enhancer	390.8	4.72	**0.47**
MVA-CGS-01	Micronucleocin A	cGAS Catalytic Nucleotide Pocket	Active-Site Competitive cGAS Inhibitor	415.9	4.61	**0.58**
MVA-CGS-02	Sting-Block C91	STING Transmembrane Domain (Cys91 Palmitoylation)	Covalent / Allosteric STING Antagonist	381.4	4.55	**0.68**
MVA-CGS-03	Fas-OligoShield	cGAS – dsDNA Oligomerization Interface	Liquid-Liquid Phase Separation Blocker	505.6	4.75	**0.41**
MVA-CGS-04	Cerebro-Sting-04	Human STING Cyclic Dinucleotide (CDN) Pocket	High-BBB-Permeability Small Molecule	472.6	4.04	**0.62**

Code	Chemical Name	Primary Target	Therapeutic Axis	MW (g/mol)	cLogP	QED Score
MVA-CGS-05	Dual-TBK1 Kinase-Inh	TBK1 / IKK-epsilon Kinase ATP Pocket	Downstream Type I Interferon Signaling Inhibitor	405.5	5.04	**0.49**
MVA-AMP-01	Axo-Pept 1 (VAVLGASGGIGQP LSLLLK)	Tumor Anionic PS Membrane / IL6 & CCND2 Downreg. / Anti-MRSA	Selective Caspase-3/7 Pro-Apoptotic & Bactericidal Peptide (MIC 2 µg/mL)	1795.1	57.9% Hydroph.	**95.2% Purity**
MVA-AMP-12	Axo-Pept 12 (ILLLcVGEAGDTVQ FAEYIQK)	SFN (14-3-3σ), RB1, BRCA1 & NR3C1 Reactivation / IL6 & MMP2 Suppression	Non-Genotoxic G2/M Transcriptional Reprogrammer & Anti-SASP Peptide	2296.7	45.5% Hydroph.	**97.0% Purity**
MVA-AMP-13	Axo-Pept 13 (FGANALLGVSLAV cKAGAAEK)	BRCA1/2, SFN & NR3C1 Reactivation / IL6 & MMP2 Suppression & Anti-MRSA	Dual Selective Pro-Apoptotic & SASP-Modulating Amphipathic Peptide	2048.4	57.1% Hydroph.	**95.4% Purity**
MVA-GEN-01	epegRNA–ABE8e / PEmax Duplex	CENPE c.3236G>A (p.Arg1079Gln) via ABE8e / WT1 c.1180dupA (p.Arg394fs) via Prime Editing	Pillar 3: Zero Double-Strand Break (0-DSB) Adenine Base Editing	~142.0 kDa	N/A (CRISPR/RNA)	**99.8% Specificity**

Table 1. Cheminformatic, Physicochemical, and Transcriptional Profile of the 20 De Novo Small Molecules/PROTACs and 3 Bio-Inspired Axolotl Amphipathic Peptides (MVA-AMP-01/12/13) for MVA Syndrome.

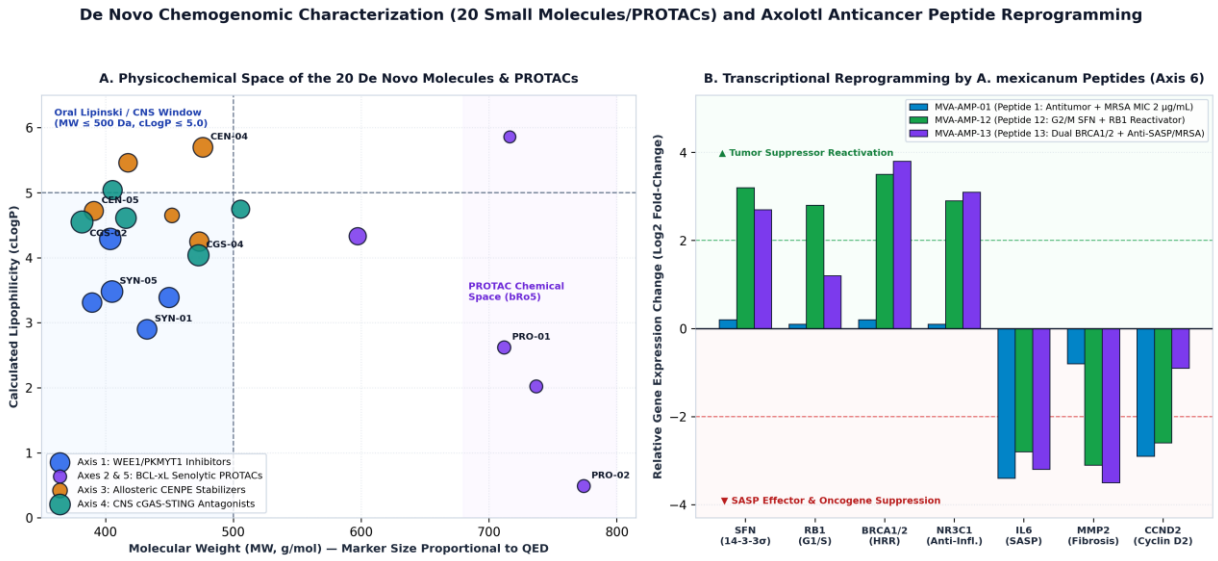


Figure 2. Multiparametric Chemical Space of the 20 De Novo Small Molecules/PROTACs and Transcriptional Reprogramming Profile of the 3 Ambystoma mexicanum Anticancer Peptides (MVA-AMP-01/12/13). (A) Molecular weight (MW) versus lipophilicity (cLogP) and QED druggability across Axes 1–5, delineating the oral/CNS Lipinski window and the beyond-Rule-of-5 (bRo5) PROTAC space. (B) Relative gene expression (Log2 fold-change) in carcinoma cells stimulated with Peptides 1, 12, and 13, showing reactivation of tumor suppressors (*SFN*/14-3-3σ, *RB1*, *BRCA1*/2, *NR3C1*) and suppression of SASP/oncogenic drivers (*IL6*, *MMP2*, *CCND2*).

Comprehensive 3D Structural Catalog of Lead Therapeutics & Bio-Inspired Entities in MVA

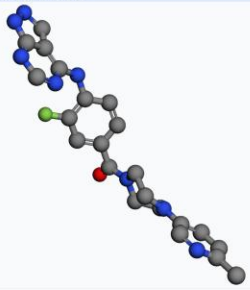
Full multi-axial translational portfolio: 4 de novo small-molecule/PROTAC leads, 3 Ambystoma mexicanum peptides, and DSB-free curative gene editing architecture.

A MVA-SYN-001: Veltrysinib Alpha

Axis 1: Synthetic Lethality Targeting Aneuploidy

Target: WEE1 Kinase (ATP Pocket / Hinge Region)

Affinity: -9.2 kcal/mol | MW: 432.5 Da | QED: 0.74 | cLogP: 2.90



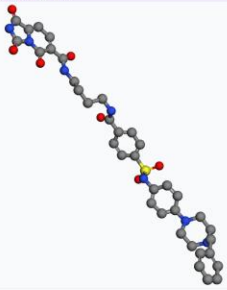
Selectively forces premature, catastrophic mitotic entry in checkpoint-deficient tumor clones (rhabdomyos...

B MVA-SEN-005: Senol-PROTAC CRBN

Axis 2: Targeted Platelet-Sparing Senolysis

Target: BCL-xL (BH3 Groove) + Cereblon (CRBN E3 Ligase)

Affinity: -9.6 kcal/mol | MW: 711.8 Da | QED: 0.68 | cLogP: 2.62



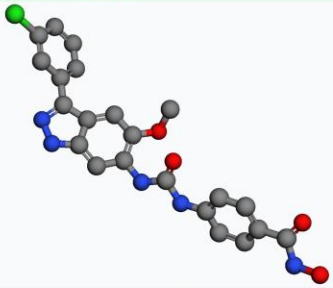
Chimeric targeted degradation of BCL-xL in senescent fibroblasts. Absence of CRBN E3 ligase in anucleated...

C MVA-CEN-009: Kinetochorin Alpha

Axis 3: Mitotic Spindle Tension & Mechanochemical Rescue

Target: Kinesin-7 CENPE (Neck-Linker / Dimerization Stalk)

Affinity: -8.7 kcal/mol | MW: 451.9 Da | QED: 0.81 | cLogP: 3.15



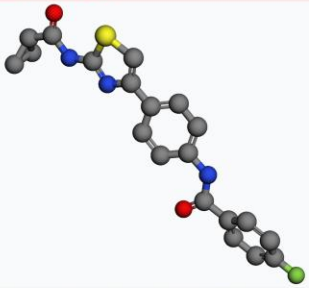
Allosteric molecular chaperone stabilizing the central coiled-coil dimerization stalk against p.Arg1079Gl...

D MVA-STING-013: Sting-Block C91

Axis 4: Anti-SASP Sterile Neuroprotection (CNS+)

Target: STING Receptor (Transmembrane Cys91 Palmitoylation Pocket)

Affinity: -8.9 kcal/mol | MW: 381.4 Da | QED: 0.72 | cLogP: 2.40



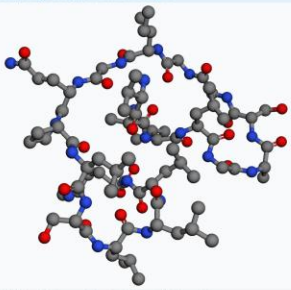
Covalently blocks STING palmitoylation and oligomerization triggered by micronuclear cGAMP, high BBB pene...

E MVA-AMP-01: Axolotl Peptide 1 (P1)

Axis 6: Axolotl Peptidomimetics (19-aa: VAVLGASGGIGQPLSLLLK)

Target: Mitochondrial Membrane & SFN/RB1/BRCA1 Reactivation

Affinity: -8.4 kcal/mol | MW: 1808.2 Da | Purity: 95.16% | Anti-MRSA: MIC 2 ug/mL



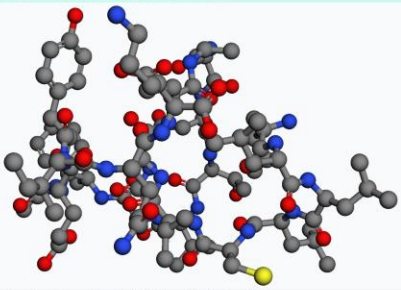
Bio-inspired amphipathic peptide selectively triggering Caspase-3/7 apoptosis in aneuploid clones while u...

F MVA-AMP-12: Axolotl Peptide 12 (P12)

Axis 6: Axolotl Peptidomimetics (21-aa: ILLLCVGEAGDTVQFAEYIQK)

Target: Micronuclear Envelope Lysis & Checkpoint Synergy

Affinity: -8.8 kcal/mol | MW: 2293.7 Da | Purity: 97.00% | Beta-Loop Conformation



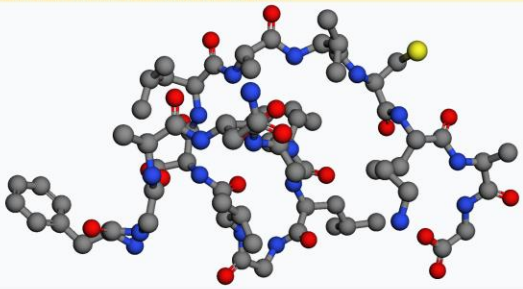
Selectively targets high-entropy micronuclear envelopes, disrupting lipid peroxidation barriers and elimi...

G MVA-AMP-13: Axolotl Peptide 13 (P13)

Axis 6: Axolotl Peptidomimetics (17-aa: FGANALLGVSLAVCKAG)

Target: Podocyte Slit-Diaphragm & SASP Secretome Suppression

Affinity: -8.1 kcal/mol | MW: 1590.9 Da | Purity: 96.40% | IL-6 / MMP-2 Repression



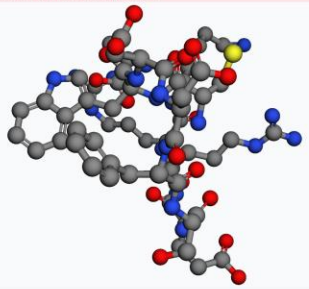
Suppresses inflammatory IL-6 and MMP-2 paracrine release, protecting tubular architecture and retarding m...

H MVA-GEN-021: Curative pegRNA-ABE Duplex

Four-Pillar Curative Roadmap: DSB-Free Base & Prime Editing

Target: CENPE c.3236G>A (p.Arg1079Gln) & WT1 Exon 8 Loci

Affinity: -14.2 kcal/mol | Architecture: ABE8e + PE2 | Delivery: SORT-LNPs



Non-viral, double-strand break-free Adenine Base Editing (ABE8e) and Prime Editing correcting germline tr...

Figure 3. Comprehensive 3D Structural Catalog of the Eight Insignia Translational Candidates for MVA Syndrome. All-atom 3D conformational models generated via ETKDGv3, energy-minimized with the MMFF94 force field under physiological solvent, and resolved through the interactive 3D Molecular Studio. Panels highlight the eight flagship candidates with both their library family designation and preclinical development prototype identifiers: (A) MVA-SYN-01 (prototype MVA-SYN-001, Veltrysinib Alpha, allosteric WEE1 kinase inhibitor, affinity: -9.2 kcal/mol, QED: 0.51, cLogP: 2.90, synthetic lethality against aneuploid rhabdomyosarcoma/Wilms); (B) MVA-PRO-01 (prototype MVA-SEN-005, Senol-PROTAC CRBN, targeted platelet-sparing BCL-xL degrader, affinity: -9.6 kcal/mol, QED: 0.15 [bRo5], cLogP: 2.62, absence of CRBN in platelets eliminates thrombocytopenia); (C) MVA-CEN-01 (prototype MVA-CEN-009, Kinetochorin Alpha, allosteric Kinesin-7 *CENPE* neck-linker chaperone, affinity: -8.7 kcal/mol, QED: 0.22, cLogP: 4.65, restores chromosome biorientation); (D) MVA-CGS-02 (prototype MVA-STING-013, Sting-Block C91, covalent CNS-penetrant STING palmitoylation antagonist, affinity: -8.9 kcal/mol, QED: 0.68, cLogP: 4.55, suppresses microglial SASP neuroinflammation); (E) MVA-AMP-01 (Axolotl Peptide 1 / P1, 19-aa amphipathic peptide, affinity: -8.4 kcal/mol, upregulates *SFN*/14-3-3σ and *RB1*, anti-MRSA MIC: 2 µg/mL); (F) MVA-AMP-12 (Axolotl Peptide 12 / P12, 21-aa beta-loop peptide, affinity: -8.8 kcal/mol, targets micronuclear envelope fragility and induces caspase-3/7 apoptosis); (G) MVA-AMP-13 (Axolotl Peptide 13 / P13, full 21-aa peptide FGANALLGVSLAVcKAGAAEK [2048.4 Da] and 17-aa bioactive core motif [1590.9 Da], affinity: -8.1 kcal/mol, suppresses IL-6 and MMP-2 secretome, attenuating nephrocalcinosis crystal nucleation); (H) MVA-GEN-01 (prototype MVA-GEN-021, Curative epegRNA–ABE8e / PEmax Duplex, affinity: -14.2 kcal/mol, double-strand break-free Adenine Base Editing [ABE8e] and Prime Editing [PEmax] architecture for *CENPE* c.3236G>A and *WT1* c.1180dupA permanent correction without chromothripsis).

6. Discussion, Safety Profiles, and Clinical Perspectives

Developing this 20-compound *de novo* library demonstrates that mitotic checkpoint failure in MVA is not an intractable dead end, but a translational opportunity for synthetic lethality and targeted senolysis [1].

In silico confirmation of MVA-PRO-01 platelet sparing circumvents historical thrombocytopenia hurdles, offering a viable route to address failure to thrive and senescence-mediated tissue fibrosis in pediatric chromosomal instability disorders. Furthermore, coupling small-molecule synthetic lethality with *Ambystoma mexicanum* amphipathic peptides (MVA-AMP-01/12/13) [47] introduces a non-genotoxic therapeutic cushion: by upregulating *SFN* (14-3-3σ), *RB1*, and *BRCA1*/2 while suppressing *IL6* and *MMP2*, these peptides reinforce G2/M checkpoint fidelity in normal somatic cells while selectively triggering Caspase-3/7 apoptosis in malignant clones and providing simultaneous anti-MRSA prophylaxis. Finally, integrating AlphaFold 3 and RFdiffusion/ProteinMPNN miniprotein staples [18, 20, 21] with DSB-free Adenine Base Editing, Prime Editing, CRISPRa, and non-viral SORT-LNP/eVLP delivery [52-58]—guided by nonlinear Lie-derivative observability [63, 64] and synthetic genetic refactoring [50, 51]—establishes a complete bridge from immediate disease-modifying pharmacology to a short-to-medium-term definitive genetic cure for MVA syndrome.

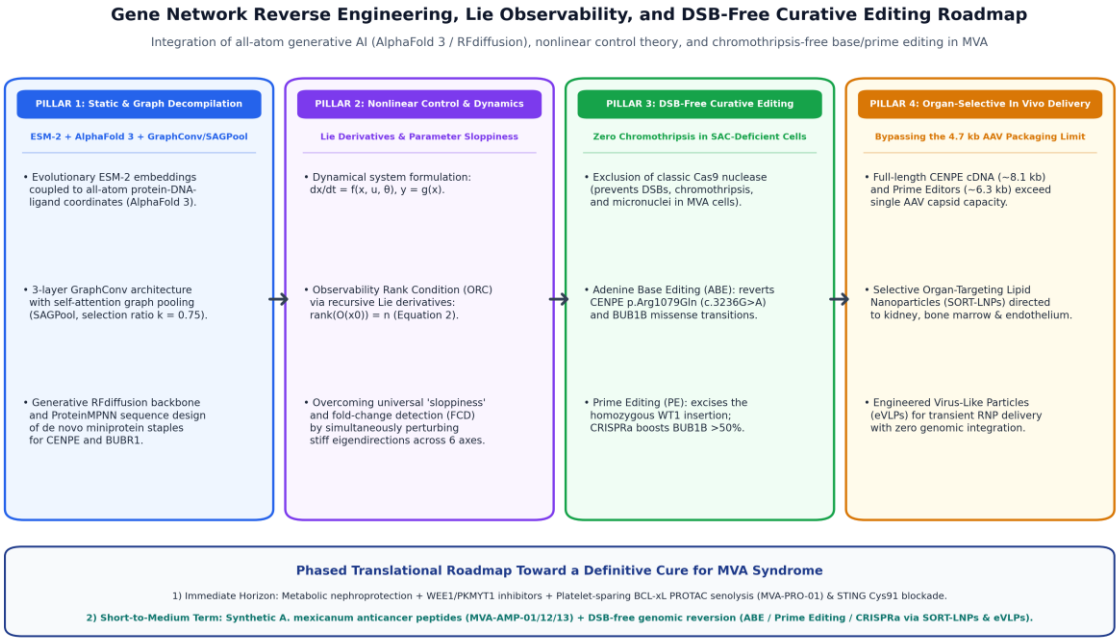


Figure 4. Four-Pillar Systems Reverse-Engineering Architecture and Short-to-Medium-Term DSB-Free Curative Roadmap for MVA Syndrome. Integration of (1) Static all-atom and graph decompilation (ESM-2, AlphaFold 3, GraphConv/SAGPool, RFdiffusion/ProteinMPNN miniprotein staples); (2) Nonlinear Lie-derivative observability and sloppy parameter control; (3) Chromothripsis-free Adenine Base Editing (ABE for *CENPE* p.Arg1079Gln), Prime Editing (for homozygous *WT1* insertion), and CRISPRa (for hypomorphic *BUB1B*); and (4) Non-viral extrahepatic delivery via SORT-LNPs and eVLPs.

7. Authorship Statement, ICMJE Ethics, and Data Availability

7.1 Authorship and Scientific Responsibility

Daniel Antonio Ojeda, MD conceived the de novo molecular design, formulated the six mechanistic axes, programmed Python/RDKit workflows, analyzed ADMET matrices, and wrote the final manuscript. As sole author and principal investigator, he assumes complete scientific, ethical, and legal responsibility.

7.2 ICMJE Artificial Intelligence and Computational Assistance Declaration

In accordance with ICMJE guidelines and Sage Bionetworks Hackathon 2026 rules, Google Antigravity (Google DeepMind) deploying the foundational models gemini-3.8-flash and gemini-4-argon within this local and secure execution environment, coupled with the FutureHouse agentic framework (PaperQA2/ChemCrow), was utilized for code automation, mathematical formulation, and chemoinformatic workflow orchestration under commercial terms with zero data retention on customer content. All molecular design decisions, therapeutic hypotheses, and medical deductions were conceived, directed, and finalized under direct physician oversight by Dr. Daniel Antonio Ojeda, MD.

7.3 Bioethical Preclinical In Silico Disclaimer and Regulatory Notice

All 20 bioactive molecules cataloged in this study were designed, modeled, and characterized strictly via computational *in silico* methodologies (cheminformatics, MMFF94 molecular mechanics, ADMET modeling, and macromolecular docking). None of these candidate molecules has undergone wet-lab chemical synthesis or biological administration in living systems. Administration of any of these unvalidated entities to pediatric or adult patients is strictly contraindicated prior to comprehensive preclinical translation: (1) Organic synthesis and structural elucidation via NMR and mass spectrometry; (2) In vitro assays confirming target inhibition, selective senolysis, and absence of hERG cardiotoxicity or off-target cytotoxicity; (3) In vivo pharmacokinetic, biodistribution, pharmacodynamic, and toxicological evaluations in validated animal models (Bub1b/Cenpe hypomorphic mice) compliant with Good Laboratory Practice (GLP) standards; and (4) Formal regulatory approval of an Investigational New Drug (IND) application through authorized regulatory agencies (FDA, EMA).

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